MAT 2040 Linear Algebra Summer 2023, CUHK(SZ)

Assignment 6

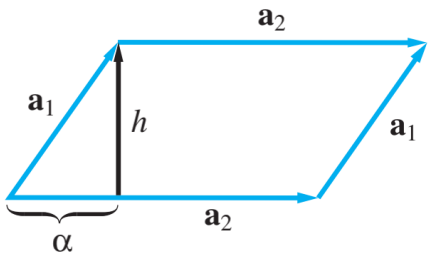
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Please note that

- **Released date**: 11th July
- **Due date**: 21th July, by 11:59 pm.
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX_ ASS7.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)
- Please make sure that your submitted file is clear and readable. Submitted file that can not be opened or not readable will get 0 point.

Question 1. Let $\boldsymbol{A}$ be a 2×2 matrix with linearly independent column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. If $\mathbf{a}_1$ and $\mathbf{a}_2$ are used to form a parallelogram P with altitude h (see the figure), show that $h^2 \|\mathbf{a}_2\|^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \left(\mathbf{a}_1^T \mathbf{a}_2\right)^2$



Question 2. Find the constants a and b such that the vectors $\mathbf{u} = \begin{bmatrix} a \\ 3 \\ -b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ 2 \\ b \end{bmatrix}$ are orthogonal and $a = b + 2$.

Question 3. State whether each of the following is true or false:

(a) If the subspaces V and W are orthogonal, then $V^{\perp}$ and $W^{\perp}$ are also orthogonal.

(b) If V is orthogonal to W and W orthogonal to Z then V is orthogonal to Z.

Question 4. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$ and let $\mathcal{S} = \text{Span}\{\mathbf{x}, \mathbf{y}\}$. Construct a matrix by $\boldsymbol{A} = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$.

(a) Show that $\boldsymbol{A}$ is symmetric.

(b) Show that $\text{Null}(\boldsymbol{A}) = \mathcal{S}^{\perp}$.

Question 5. Let $\mathbf{x} \in \mathbb{R}^n$, $\|\mathbf{x}\|_\infty = \max_{i=1,\cdots,n} |\mathbf{x}_i|$ and $\|\mathbf{x}\|_p = (\sum_{i=1}^n |\mathbf{x}_i|^p)^{1/p}$.

(a) Show that $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2$.

(b) Show that $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$.

Question 6. An $n \times n$ matrix $\boldsymbol{Q}$ is said to be an orthogonal matrix if the column vectors of $\boldsymbol{Q}$ form an orthonormal set in $\mathbb{R}^n$. Find a, b, c, d such that

$$\boldsymbol{Q} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} & a \\ \frac{1}{2} & -\frac{1}{2} & 0 & b \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} & c \\ \frac{1}{2} & -\frac{1}{2} & 0 & d \end{bmatrix}$$

is an orthogonal matrix.

Question 7. Find the angle between the vectors $\mathbf{v}$ and $\mathbf{w}$:

$$\mathbf{v} = (2, 1, 3)^T, \ \mathbf{w} = (12, 6, 18)^T$$

Question 8. For the following pair of vectors $\mathbf{x}$ and $\mathbf{y}$, find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ and verify that $\mathbf{p}$ and $\mathbf{x} - \mathbf{p}$ are orthogonal:

$$\mathbf{x} = (3, 5)^T, \ \mathbf{y} = (1, 1)^T$$

Question 9. If $\mathbf{x} = (x_1, x_2)^T, \mathbf{y} = (y_1, y_2)^T$, and $\mathbf{z} = (z_1, z_2)^T$ are arbitrary vectors in $\mathbb{R}^2$, prove that:

(a) $\mathbf{x}^T\mathbf{x} \geq 0$

(b) $\mathbf{x}^T\mathbf{y} = \mathbf{y}^T\mathbf{x}$

(c) $\mathbf{x}^T(\mathbf{y} + \mathbf{z}) = \mathbf{x}^T\mathbf{y} + \mathbf{x}^T\mathbf{z}$

Question 10. (a) Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{x} = (x_1, x_2, x_3)^T$ and $\mathbf{y} = (y_1, y_2, y_3)^T$. Let

$$\boldsymbol{A} = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{pmatrix}.$$

Show that $S^{\perp} = \text{Null}(\boldsymbol{A})$.

(b) Find the orthogonal complement of the subspace of $\mathbb{R}^3$ spanned by $(-1, 2, 1)^T$ and $(1, -1, 0)^T$.

Question 11. For each of the following systems $\boldsymbol{A}\mathbf{x} = \mathbf{b}$, find all least squares solutions:

(a) $\boldsymbol{A} = \begin{pmatrix} 1 & 4 \\ 2 & 1 \\ -1 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

(b) $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix}$

Question 12. Let $\boldsymbol{P} = \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T$, where $\boldsymbol{A}$ is an $m \times n$ matrix of rank n.

(a) Show that $\boldsymbol{P}^2 = \boldsymbol{P}$.

(b) Prove that $\boldsymbol{P}^k = \boldsymbol{P}$ for $k = 1, 2, \ldots$

(c) Show that $\boldsymbol{P}$ is symmetric. [Hint: If $\boldsymbol{B}$ is non-singular, then $\left(\boldsymbol{B}^{-1}\right)^T = \left(\boldsymbol{B}^T\right)^{-1}$.]

Question 13. Let $\mathbf{x} = (1, 2, 0, 1)^T$ and $\mathbf{y} = (4, 1, 2, 0)^T$.

(a) Determine the angle θ between $\mathbf{x}$ and $\mathbf{y}$.

(b) Find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$.

(c) Verify that $\mathbf{x} - \mathbf{p}$ is orthogonal to $\mathbf{p}$.

(d) Compute $\|\mathbf{x} - \mathbf{p}\|_2, \|\mathbf{p}\|_2, \|\mathbf{x}\|_2$ and verify that the Pythagorean law is satisfied.

Question 14. Let

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

(a) Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $\mathbf{x} = (1, 1, 1)^T$. Write $\mathbf{x}$ as a linear combination of $\mathbf{u}_1, \mathbf{u}_2$, and $\mathbf{u}_3$ using Theorem 21.13 and use Parseval's formula to compute $\|\mathbf{x}\|$.

Question 15. Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ of Question 14. Let $\mathbf{x} = (1, 2, 2)^T$. Find the projection $\mathbf{p}$ of $\mathbf{x}$ onto S. Show that $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_2$ and $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_3$.

Question 16. Let $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for an inner product space V. If $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$ is a vector with the properties $\|\mathbf{x}\| = 5, \langle \mathbf{u}_1, \mathbf{x} \rangle = 4$, and $\mathbf{x} \perp \mathbf{u}_2$, then what are the possible values of c_1, c_2, c_3?

Question 17. Given the basis $\left\{(1,2,-2)^T,(4,3,2)^T,(1,2,1)^T\right\}$ for $\mathbb{R}^3$, use the Gram-Schmidt process to obtain an orthonormal basis.

Question 18. Let $\boldsymbol{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 12 \\ 6 \\ 18 \end{pmatrix}$

(a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of $\boldsymbol{A}$.

(b) Factor $\boldsymbol{A}$ into a product $\boldsymbol{QR}$, where $\boldsymbol{Q}$ has an orthonormal set of column vectors and $\boldsymbol{R}$ is upper triangular.

(c) Solve the least squares problem $A\mathbf{x} = \mathbf{b}$.